Studying temperature dependence of resistivity of semiconductor crystal and estimating its band-gap using the Four Probe method

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Contents

1 Abstract **1**

2 Theory **2**

3 Observations & Data Analysis **3**
3.1 Plots 5
3.2 Curve Fitting 6
3.2.1 Heating Plot 6
3.2.2 Cooling plot 6
3.3 Calculation 7
3.3.1 Correction Factor Calculation 7

4 Error Analysis **7**
4.1 Analysis of Heating-Cooling Discrepancy 8

5 Results and Precautions **8**

1 Abstract

This lab experiment investigates the temperature dependence of resistivity in a semiconductor crystal using the Four Probe method. By measuring the resistivity at different temperatures, we can estimate the band gap of the semiconductor material. The experiment involved a carefully designed setup to ensure accurate measurements, and the results indicated a clear relationship between temperature and resistivity, allowing us to estimate the band gap.

2 Theory

Intrinsic semiconduction is the process in which thermally or optically excited electrons contribute to conduction. In the absence of photonic excitation, intrinsic semiconduction occurs at temperatures above 0 K, as sufficient thermal agitation is required to transfer electrons from the valence band to the conduction band.

The total electrical conductivity (σ) for intrinsic semiconduction is the sum of the conductivities of the valence and conduction band carriers, which are holes and electrons, respectively. It can be expressed as:

$$\sigma = e(n_e\mu_e + n_h\mu_h)$$

Here, n_e and μ_e are the electron's concentration and mobility, and n_h and μ_h are the hole's concentration and mobility.

Mobility (μ) determines the average drift velocity in the presence of an applied external field. It also depends on the temperature. Mobility is a quantity that directly relates the drift velocity (v_d) of charge carriers to the applied electric field (E) across the material:

$$\mu = \frac{v_d}{E}$$

In the intrinsic region, the number of electrons is equal to the number of holes ($n_e = n_h = n_i$), so this equation implies that:

$$\sigma = en_i(\mu_e + \mu_h)$$

Four probes are placed on the surface of the sample. If the separation between each of the adjacent probes is x, then, if current passes through the outer two probes and resistivity is measured across the inner two probes, then the potential difference across the inner two probes is given by:

$$V = \frac{\rho I}{2\pi x} \tag{1}$$

So, the resistivity of the sample is:

$$\rho = \frac{V}{I} \times 2\pi x$$

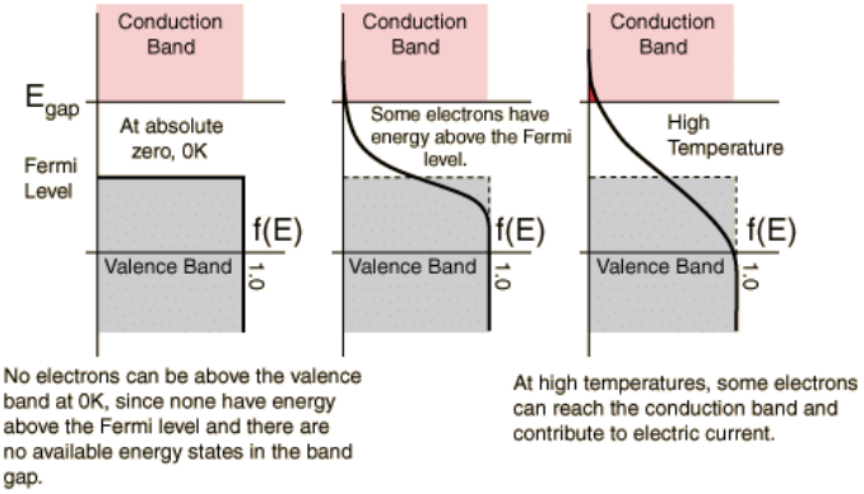


Figure 1: Band diagram of a semiconductor

Now, since the thin slice is placed on a non-conducting bottom surface, we have to introduce a correction factor, and the corrected resistivity is given by:

$$\rho_c = \frac{\rho}{f_2\left(\frac{w}{x}\right)} \tag{2}$$

where "w" is the thickness of the sample and "x" is probe spacing. The function $f_2\left(\frac{w}{x}\right)$ is given by:

$$f_2\left(\frac{w}{x}\right) = 1 + 4\frac{x}{w}\sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{\left(\frac{x}{w}\right)^2 + n^2}} - \frac{1}{\sqrt{4\left(\frac{x}{w}\right)^2 + (2n)^2}}\right) \tag{3}$$

For semiconductors, if ρ is the resistivity value, then:

$$\rho_c = A\exp\left(\frac{E_g}{2k_BT}\right)$$

where A is a constant, E_g is the band gap energy, k_B is the Boltzmann constant, and T is the absolute temperature.

$$\ln\rho_c = \ln A + \frac{E_g}{2k_BT} \tag{4}$$

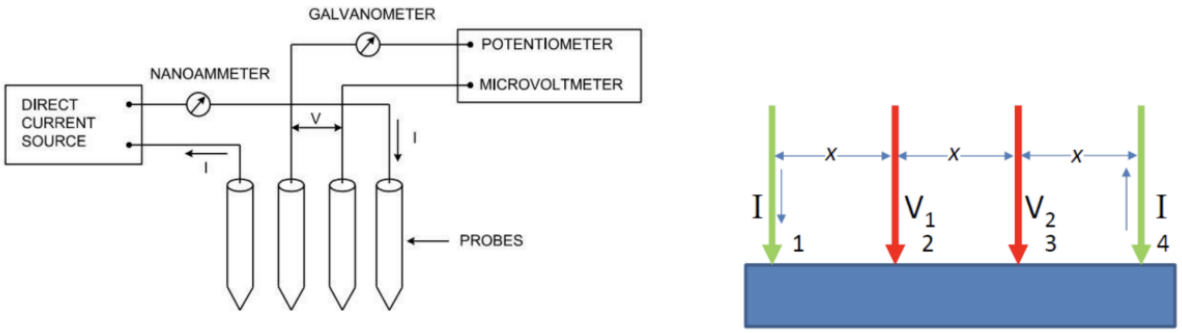


Figure 2: Four probes setup

3 Observations & Data Analysis

Our apparatus has a sample thickness (w) of 0.5 mm and a probe separation (x) of 2 mm. It is crucial to acknowledge that fluctuations occurred in the current generated by the constant current source at the onset and conclusion of the heating phase due to instrumental error. These variations will introduce non-linearity in the plot of $\ln\rho_c$ against $\frac{1}{T}$. However, these fluctuations will be considered and corrected during the data analysis.

Temperature($^\circ C$)	**voltage(V)**	$\boldsymbol{\rho_c(\Omega \cdot m)}$	$\boldsymbol{1/T(K^{-1})}$
26	0.396	-1.7076	0.003343
29	0.392	-1.71775	0.00331
32	0.384	-1.73837	0.003277
35	0.376	-1.75943	0.003245
38	0.364	-1.79186	0.003214
41	0.353	-1.82255	0.003183
44	0.34	-1.86007	0.003153
47	0.325	-1.90519	0.003124
50	0.308	-1.95892	0.003095
53	0.293	-2.00884	0.003066
56	0.272	-2.08321	0.003038
59	0.259	-2.13219	0.003011
62	0.244	-2.19185	0.002984
65	0.227	-2.26407	0.002957
68	0.213	-2.32772	0.002931
71	0.198	-2.40075	0.002906
74	0.184	-2.47408	0.002881
77	0.171	-2.54735	0.002856
80	0.159	-2.62011	0.002832
83	0.146	-2.70541	0.002808
86	0.136	-2.77636	0.002784
89	0.126	-2.85273	0.002761
92	0.118	-2.91833	0.002739
95	0.108	-3.00688	0.002716
98	0.1	-3.08385	0.002694
101	0.093	-3.15642	0.002673
104	0.085	-3.24636	0.002651
107	0.08	-3.30699	0.002631
110	0.074	-3.38495	0.00261
113	0.07	-3.44052	0.00259
116	0.065	-3.51463	0.00257
119	0.06	-3.59467	0.00255
122	0.058	-3.62857	0.002531
125	0.055	-3.68168	0.002512
128	0.0515	-3.74743	0.002493
131	0.0495	-3.78704	0.002474
134	0.046	-3.86037	0.002456
137	0.0428	-3.93248	0.002438
140	0.0398	-4.00515	0.00242

Table 1: Heating data with Current(I) maintained at 4.95mA

Temperature($^\circ C$)	**voltage(V)**	$\rho_c(\Omega \cdot m)$	$1/T(K^{-1})$
135	0.0467	-3.84527	0.00245
132	0.0504	-3.76902	0.002468
129	0.0535	-3.70933	0.002487
126	0.0584	-3.6217	0.002505
123	0.0616	-3.56835	0.002524
120	0.0668	-3.48731	0.002544
117	0.0713	-3.42212	0.002563
114	0.0769	-3.34651	0.002583
111	0.0825	-3.27622	0.002603
108	0.0889	-3.2015	0.002624
105	0.0968	-3.11637	0.002644
102	0.1046	-3.03887	0.002666
99	0.1126	-2.96517	0.002687
96	0.1217	-2.88746	0.002709
93	0.1314	-2.81077	0.002731
90	0.1428	-2.72757	0.002754
87	0.1549	-2.64624	0.002777
84	0.1675	-2.56803	0.0028
81	0.1815	-2.48776	0.002824
78	0.1954	-2.41397	0.002848
75	0.21	-2.34191	0.002872
72	0.223	-2.28184	0.002897
69	0.241	-2.20422	0.002923
66	0.257	-2.13994	0.002949
63	0.276	-2.06861	0.002975
60	0.292	-2.01226	0.003002
57	0.31	-1.95244	0.003029
54	0.328	-1.896	0.003057
51	0.345	-1.84547	0.003085
48	0.357	-1.81128	0.003114

Table 2: Cooling Data with Current(I) maintained at 4.95mA

3.1 Plots

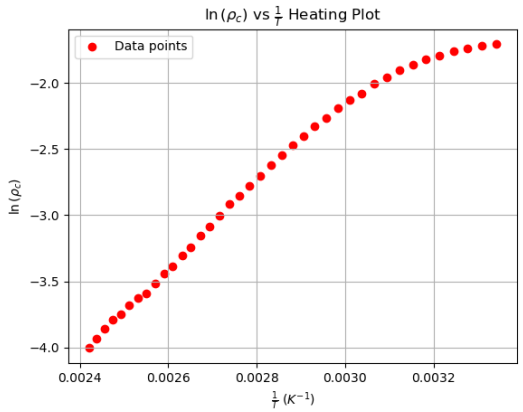


Figure 3: Heating plot

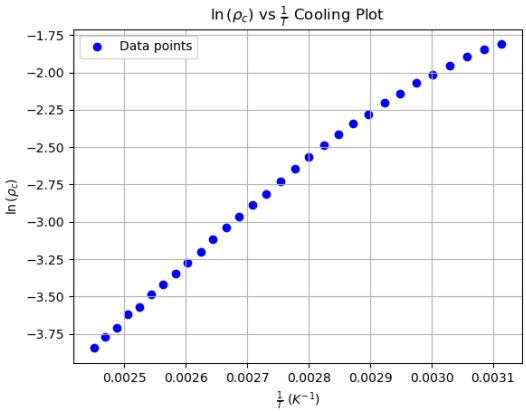
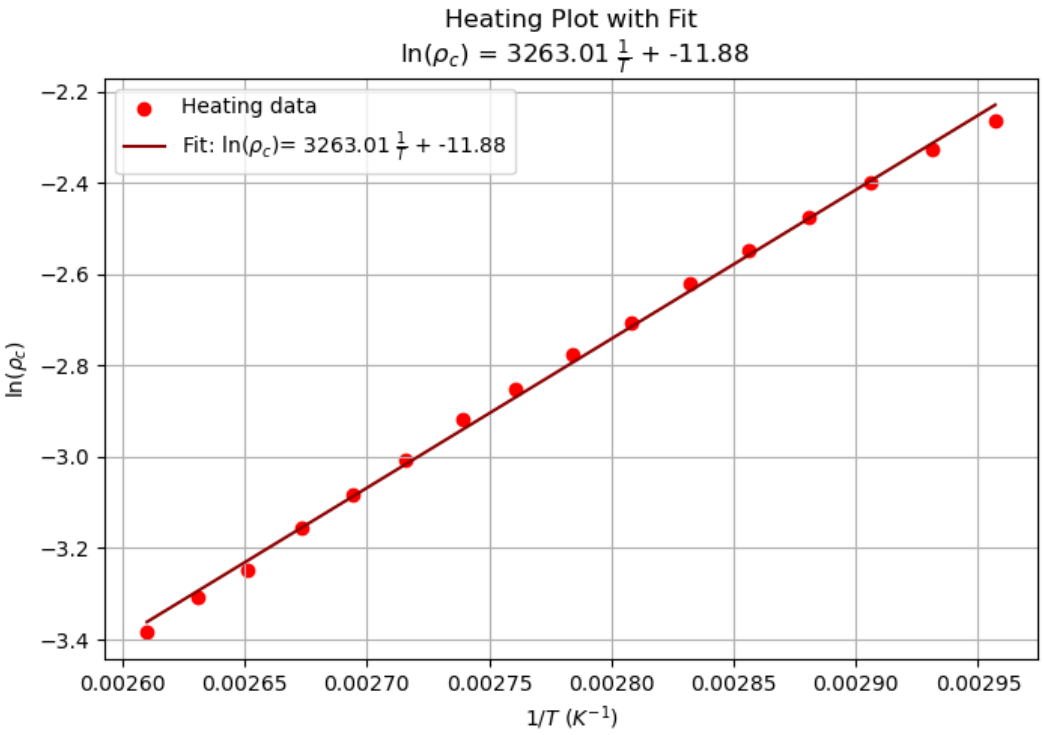


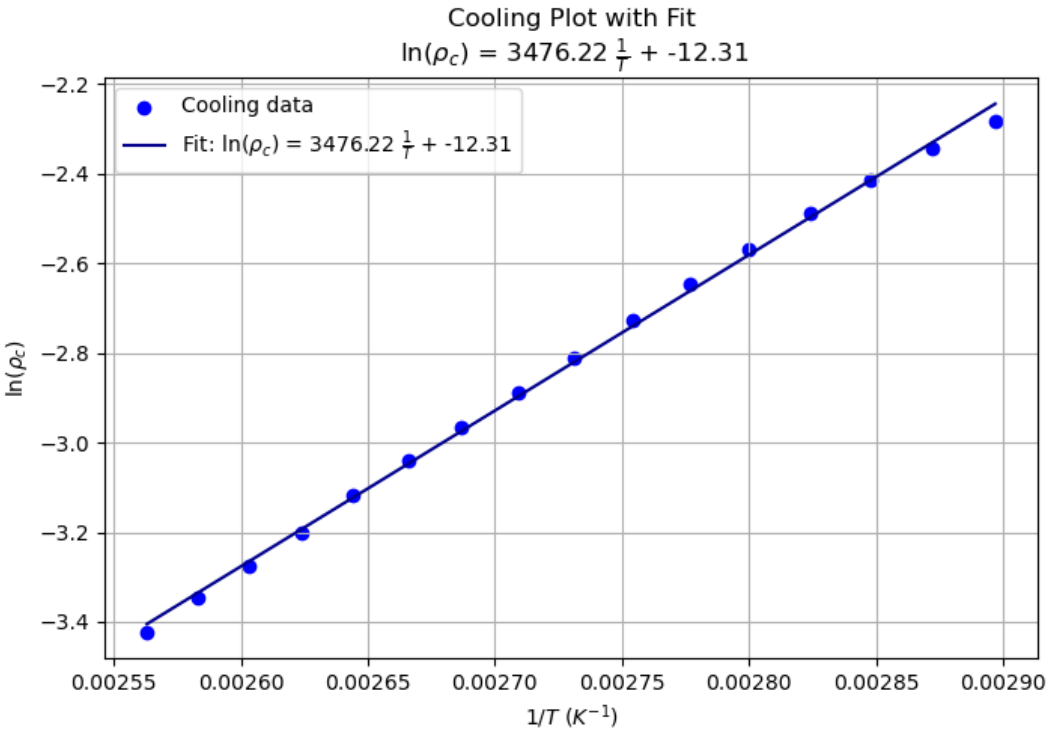
Figure 4: Cooling Plot

3.2 Curve Fitting

3.2.1 Heating Plot



3.2.2 Cooling plot



3.3 Calculation

3.3.1 Correction Factor Calculation

For $\frac{w}{x} = 0.25$, the correction factor $f_2\left(\frac{w}{x}\right)$ is computed from the infinite sum:

$$f_2\left(\frac{w}{x}\right) = 1 + 4\frac{x}{w}\sum_{n=1}^{\infty}\left[\frac{1}{\sqrt{\left(\frac{x}{w}\right)^2 + n^2}} - \frac{1}{\sqrt{4\left(\frac{x}{w}\right)^2 + (2n)^2}}\right]$$

We can the below approximation for small values of W/S to compute f_2

$$f_2\left(\frac{W}{S}\right) = \frac{2S}{W}\log_e 2$$

With $\frac{x}{w} = 4$, numerical evaluation (truncating at $n = 10$) yields $f_2(0.25) \approx 5.545$, confirming the value used in resistivity corrections.

We have used ρ_c in units of $\Omega\cdot$ m (i.e., SI units) in our computations and plots.

So,

$$Slope = \frac{E_g}{2K_B}$$

$$E_g = 2K_B \times Slope \tag{4}$$

we know $K_B = 8.617 \times 10^{-5} eV/K$. From eqn (4), we obtain-

$$\boxed{\boldsymbol{E_g^{\text{cooling}} = 0.6\,\text{eV}}} \tag{1}$$

$$\boxed{\boldsymbol{E_g^{\text{heating}} = 0.562\,\text{eV}}} \tag{2}$$

then we can say,

$$\boxed{\boldsymbol{E_g^{\text{average}} = 0.581\,\text{eV}}} \tag{3}$$

4 Error Analysis

Assuming that we have no error associated with probe spacing calculation, we can calculate the relative error in resistivity as follows:

$$\frac{\delta\rho_c}{\rho_c} = \frac{\delta V}{V} + \frac{\delta I}{I}$$

In our case, due to a faulty constant current source instrument, δI was about 0.01 mA due to fluctuations in current, although I was 4.95 mA for most of the time. For the voltage readings, δV was 0.001 V. Now, considering the readings recorded while cooling and taking the voltage reading of 0.039 V (to get an idea of the upper bound of the error), we have:

$$\frac{\delta\rho_c}{\rho_c} = \frac{0.001}{0.039} + \frac{0.01}{4.95} \approx 0.0277$$

Now, from equation (4), we can write by taking differentials:

$$\frac{\delta\rho_c}{\rho_c} = \frac{T\delta E_g - E_g\delta T}{2k_BT^2} = \frac{1}{2k_BT}\left(\delta E_g - E_g\frac{\delta T}{T}\right)$$

and thus we have,

$$\frac{\delta E_g}{E_g} = \frac{2k_BT}{E_g}\left(\frac{\delta\rho_c}{\rho_c}\right) + \frac{\delta T}{T}$$

In this case, δT is 1 K (the least count of the temperature sensor), and the average value of temperature can be taken around 363 K. Also, we know that $k_B = 8.617 \times 10^{-5}$ eV/K, and hence:

$$\frac{\delta E_g}{E_g} = \frac{2 \times 8.617 \times 10^{-5} \times 363}{0.581} \times 0.0334 + \frac{1}{363} \approx 0.0057$$

Thus, the error in band gap calculation is $\delta E_g \approx 0.0033$ eV.

4.1 Analysis of Heating-Cooling Discrepancy

The band gap estimates differ between heating ($E_g = 0.562$ eV) and cooling ($E_g = 0.6$ eV), resulting in a 6.8% variation. This could arise from thermal hysteresis in the sample, slight mobility changes not accounted for in the model, or instrumental artifacts like probe contact resistance varying with temperature. Future experiments could minimize this by slower temperature ramps.

5 Results and Precautions

From the data analysis, we can conclude that:

- The resistivity of the given semiconducting sample was measured, and the data clearly showed a decrease in resistivity with increasing temperature.
- The band gap of the given semiconducting sample was measured to be $E_g = (0.581 \pm 0.0033)$ eV
- Material context The estimated band gap of 0.581 ± 0.0033 eV is consistent with germanium, which has a literature value of approximately 0.66 eV at 300 K, with the difference attributable to temperature dependence or sample impurities.

While performing the experiment, the following precautions should be kept in mind:

- The current should be monitored, as slight variations in the current can lead to significant measurement errors.
- The knob on the micro-voltmeter should be adjusted to set the offset voltage to 0.
- The voltage varies occasionally, so while taking the reading we should wait until it becomes stable.
- While taking the values of decreasing voltage (while heating), it is better to maintain a 2-3 degree Celsius interval between each measurement because the temperature increases so rapidly that it is hard to keep an eye on that.